Filippo Giovagnini

Mathematics PhD @ ICL
Data Scientist @ SP

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LinkedIn Website GitHub
London, UK

Research Profile

Mathematics PhD candidate at Imperial College London working on deep-learning theory, high-dimensional probability, and stochastic analysis, alongside applied ML experience in NLP, transformers, model evaluation, and unstructured-data pipelines. Strong Python and PyTorch experience, with research writing and end-to-end ownership in agentic ML work.

For more details about my research, projects, and publications, visit https://filippogiovagnini.github.io/.

Selected Machine Learning Work

- **Kaggle Silver Medal – NeuroGolf 2026.** Ranked **40th** (**top 1.4%**) in an AI-agent and ONNX optimization challenge. Built an agent workflow to refine compact ONNX solvers under exact-output, parameter, and memory constraints. Code; Competition.
- **Decoder-only transformer from scratch.** Implemented causal attention, residual blocks, normalization, and a language-model head in PyTorch; developing the TinyStories training and evaluation pipeline. Ongoing.

Machine Learning Experience

Stripe Partners — London, UK
Data Scientist, Part-time — Mar 2026 – Present

- Develop transformer-based statistical pipelines and model-ready features for customer analyses, including PayPal/Venmo and Novo Nordisk.
- Contribute to large-scale conversation-data research; see our latest work on conversational AI in financial judgement and decision-making here.

Aleta Index — London, UK
Machine Learning Engineer, Part-time — Feb 2025 – Feb 2026

- Built real-time NLP pipelines for topic, sentiment, and framing analysis of financial-news streams.

Selected Research Publications

Go to my website for a complete list.

- **Universality in Deep Neural Networks: An approach via the Lindeberg exchange principle**. With S. Kotitsas and M. Romito, 2026.
- **Non-selection of Lagrangian trajectories in the zero-noise limit for a class of stochastic regularizations**. With L. Galeati and M. Sorella, 2026.
- **A uniform particle approximation to the Navier–Stokes–alpha models in three dimensions with advection noise**. With D. Crisan, 2025.

Education

Imperial College London — PhD in Mathematics – Stochastic Analysis, MFC CDT — London, UK; Dec 2023 – Present

University of Pisa — MSc in Mathematics – 110/110 cum laude, GPA 30/30 — Pisa, Italy; 2023

ETH Zürich — Exchange Programme – Quantitative Risk Management and Optimal Transport — Zürich, Switzerland; 2023

University of Pisa — Bsc in Mathematics – 110/110 cum laude, GPA 29.85/30 — Pisa, Italy; 2023

Technical Skills

Programming & frameworks: Python, PyTorch, Scikit-learn, NumPy, Pandas, Polars, Git, Linux, C++.

Machine learning: Transformers, NLP, embeddings, agentic workflows, model compression, evaluation, probabilistic modelling, feature engineering, real-time and unstructured-data pipelines.

Research methods: deep-learning theory, high-dimensional probability, stochastic calculus, SDEs/SPDEs, Monte Carlo methods, numerical analysis, statistical learning, optimal transport.

Additional: TA for MSc Stochastic Calculus for Finance at Imperial College Business School (2023–Present); Italian (native), English (fluent), Spanish (conversational).